Hardened concrete: strength, behaviour and testing

Notes from the lecture slides “Module 7 – Hardened Concrete”

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1 What this module covers

Hardened concrete is judged mainly by its **mechanical properties**. The module covers:

- the factors that affect strength,
- **compressive strength** (the property most used in design, e.g. for **columns**),
- **tensile strength**,
- **flexural strength** (important for **beams** and pavements),
- **fatigue strength**, and
- **bond strength** with steel,

plus how concrete behaves under load and two **non-destructive tests** (UPV and rebound hammer).

The last slide places these in the overall picture of **material characterization**:

Fresh concrete properties	Shrinkage cracking properties	Hardened concrete properties
Slump cone, V-B consistency, inverted slump cone	**Early age**: plastic shrinkage. **Long term**: free shrinkage, restrained shrinkage, residual stresses	**Mechanical**: strength, toughness, impact resistance. **Durability**: cracked permeability

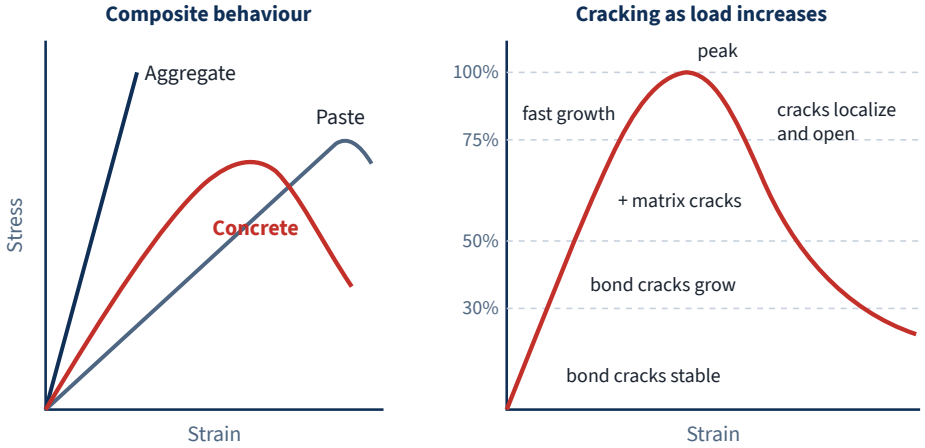
2 Concrete as a composite material

2.1 A multiphase material at every scale

Scale	What concrete looks like
Macroscopic	**Coarse aggregate embedded in cement mortar**
Millimetric	Mortar = **sand embedded in hydrated cement paste**
Microscopic	Hydrated paste = **C-S-H, $Ca(OH)_2$, pores and unhydrated cement**
Nanometric	C-S-H = **poorly crystallized agglomerations of particles, with gel pores**

Behaviour is studied with a **multi-scale approach**: the **structural scale** (e.g. a bridge), the **material macro- and millimetric scales** (a core showing aggregate and mortar), and the **microscopic scale** (SEM images of hydration products). What happens at a smaller scale explains behaviour at the larger one.

2.2 Composite behaviour



Left: aggregate and cement paste on their own are nearly linear up to failure, but concrete is not. Right: cracking stages along the stress–strain curve of concrete in compression.

- **Aggregate** alone: stiff and almost **linear** up to failure.
- **Cement paste** alone: less stiff, but also nearly **linear** to failure.
- **Concrete**: its stiffness lies **between** the two, yet its curve is **non-linear** and shows **softening** after the peak.

Why? Because concrete is not just the sum of its parts. **Microcracks already exist at the interfacial transition zone (ITZ)** between aggregate and paste **even before loading** (from shrinkage and bleeding). Under load these cracks grow, and that makes the curve bend.

2.3 Elastic properties

Property	Definition	Measured under	Typical value
Elastic modulus, E	Ratio of **stress to strain** ($E = \sigma/\varepsilon$)	Uniaxial compression	**30–50 GPa**
Poisson's ratio, ν	Ratio of the magnitudes of **transverse (lateral) to axial strain**	Uniaxial compression	**0.18–0.24**

For design, **IS 456** gives the short-term static modulus as $\mathbf{E_c = 5000\sqrt{f_{ck}}}$ (MPa). For example, M30 → 5000 × √30 = **27,400 MPa ≈ 27 GPa**.

2.4 Cement–aggregate bond

The composite response also depends on the **bond between paste and aggregate**. Factors that affect it:

- **Shape and texture of coarse aggregate.** Rough aggregate gives better mechanical interlock and adhesion: **flexural and tensile strengths can be up to 30% higher** with rough aggregate than with smooth aggregate.
- **Vibration (compaction). Air voids weaken the interface**, so good vibration improves bond.
- **Water absorption by the aggregate** improves bond; **porous aggregates bond very well** with mortar (they draw paste into their surface pores).
- **Chemical reaction**: some aggregates, such as **limestone**, may react with the cement, giving **better bond**.

2.5 Effect of aggregate type

Stress–strain curves of 100 × 200 mm cylinders at 28 days made with **diabase, limestone, gravel and granite** aggregate show that the **range of compressive strength (f'c)** between aggregate types is wider than the range of **elastic modulus (E)**.

Aggregate type has a larger influence on compressive strength than on elastic modulus, especially in high-strength concrete (where the paste is so strong that the aggregate can become the weak link).

3 Factors affecting the strength of concrete

3.1 Material and curing factors

Factor	Effect
Water/cement ratio	The most important factor. **Abrams' law**: $f_c = A / B^{w/c}$, i.e. **strength falls as w/c rises**, for fully compacted concrete

Factor	Effect
Gel/space ratio (Powers)	Strength depends on how much of the available space is filled by hydration products: $f_c = 234\,x^3$ MPa, where $x = 0.657\alpha / (0.319\alpha + w/c)$ and α = degree of hydration
Degree of compaction	Voids reduce strength sharply (Module 6: 5% voids → about 30% less strength)
Age and degree of hydration	Strength keeps rising as hydration continues
Curing (moisture and temperature)	Moist curing keeps hydration going; drying stops it
Aggregate (strength, shape, texture, max size) and **bond**	See section 2.4 and 2.5
Cement type and admixtures	Rate of strength gain; mineral admixtures add later strength

Worked example: gel/space ratio

For fully hydrated paste ($\alpha = 1$) at w/c = 0.5: $x = 0.657 / (0.319 + 0.5) = 0.802$, so $f_c = 234 \times 0.802^3 =$ **121 MPa** (intrinsic paste strength). At w/c = 0.4, $x = 0.914$ and $f_c =$ **179 MPa**. Lower w/c → more of the space filled with gel → stronger.

3.2 Test factors that change the measured compressive strength

The **same concrete** can give different "strengths" depending on how it is tested.

1. End restraint between specimen and loading platen

- The steel platens are much stiffer than concrete and expand less sideways. **Friction** at the contact surfaces stops the ends of the specimen from expanding laterally.
- This creates a **lateral compressive stress** near the ends, so an element near the end is under **triaxial compression**, which is stronger than uniaxial.
- **More friction on the loading surfaces → higher maximum load.** This is why cubes (h/d = 1) give higher strength than cylinders (h/d = 2).

2. Specimen dimensions (slenderness)

- **Increasing the slenderness (height/diameter ratio) decreases the maximum stress.**
- Taking h/d = 2 as the reference (relative strength 1.0): at h/d = 1.0 the relative strength is about 1.1–1.25, and at h/d = 0.5 it can be 1.4–1.8 or more.
- The effect is **smaller for higher-strength concrete** (curves shown for 14, 26 and 52 MPa).
- For the same concrete, **cylinder strength ≈ 0.8 × cube strength**.

3. Loading rate

- **A higher loading rate gives higher strength and a slightly higher elastic modulus.**
- Taking strength at 0.2 MPa/s as 100%, very slow loading (10^{-4} MPa/s) gives about 85%, while impact-type rates (10^{5} MPa/s) can give 180–200%.
- At **high rates** the concrete appears **stiffer** and fails **suddenly**; at **slow rates** it is less stiff, deforms more, and fails **gradually**, showing more ductility (toughness).
- Standards therefore fix the loading rate (IS 516: 14 N/mm^2 per minute, i.e. about 0.23 MPa/s, for cubes).

4. Loading surface of the specimen

- The loaded faces must be **as parallel and as plane as possible**.
- Use **good-quality, rigid moulds**.
- **Cylinders** (cast vertically, so the top face is rough) need **capping** with sulphur mortar or similar to get a plane top. The **capping must not fail before the specimen**. For **high-strength concrete**, cylinder ends may need to be **ground** flat instead.
- Cubes are tested **on their side**, so the two loaded faces are the smooth faces that were against the mould.

3.3 Satisfactory failure modes of cubes (BS EN 12390-3)

	Non-explosive (satisfactory)	Explosive (also satisfactory)
Shape after failure	**Hourglass**: all four exposed faces cracked about equally, leaving two **pyramids** (cones) joined at the tips	Breaks **violently** into a single **pyramid** shape
Why	**End restraint** confines the top and bottom (lateral compressive stress from platen friction); the middle, free to expand, fails first	Stored strain energy is released suddenly: common with **high-strength concrete** or a **less rigid testing machine**

Unsatisfactory failures (e.g. a single vertical crack on one face, or tensile splitting) indicate a problem with the **testing machine (not rigid enough), alignment, or the specimen**, and should be reported.

4 Stress–strain behaviour in compression

4.1 Cracking stages

Under uniaxial compression, the stress–strain curve can be divided according to how cracks develop:

Stress (% of peak)	Cracking	Curve
0–30%	**Negligible**: pre-existing **bond (ITZ) cracks** are stable	**Linear**
30–50%	**Slow growth of bond cracks**	Starts to bend slightly
50–75%	**Slow growth of matrix (mortar) cracks** + bond cracks; cracks start entering the mortar	Clearly non-linear
75–100%	**Fast growth of matrix cracks**; cracks link up into continuous patterns	Strongly curved; volume starts to increase
After the peak	**Cracks localize** into a band and **open**; failure by sliding along these cracks	**Falling (descending) branch**

After the peak, the curve has a **falling branch** (crack propagation) and then a flatter **converging branch**, where the load is carried by **friction and interlocking**

between cracked sections, out to about 0.004 strain and beyond.

4.2 Constitutive relationship (summary)

- The curve is **linear up to about 40% of the ultimate load**, then non-linear.
- The **non-linearity is due to the heterogeneity** of concrete: the **ITZ and microcracks**.
- **Strain at ultimate (peak) stress ≈ 0.003** for normal-strength concrete. (IS 456 uses 0.002 at peak and 0.0035 as the ultimate strain in design.)
- **Behaviour in tension is similar** in shape, but at much lower stress.

4.3 Localization

- **Before the peak**, microcracks are **spread through** the whole specimen.
- **After the peak**, damage **localizes** into a narrow **band**. That band keeps softening (strain-softening, load falling), while the material outside it **unloads**.
- So after the peak, the measured "strain" depends on the **gauge length**: the post-peak curve is a property of the specimen and the test, not purely of the material.

4.4 Effect of strength level

As concrete strength increases (curves from about 20 to 80 MPa):

- a slightly **higher elastic modulus** (stiffer),
- a **longer linear portion**,
- the strain at peak rises a little (about 0.002 to 0.003), and
- a **sharper, steeper post-peak descent**: higher-strength concrete is **more brittle** and fails more suddenly.

4.5 Poisson's ratio and volume change

- At low stress, lateral strain is small and $\nu \approx 0.18–0.24$.
- **Beyond a certain axial stress (σ_A), the transverse strain increases considerably**, causing an **apparent increase in Poisson's ratio** (because of vertical cracks opening).

- **Volumetric strain** = $\varepsilon_1 + 2\varepsilon_3$ (axial + 2 × lateral). At first the volume **decreases** (compression); near about 75–90% of peak, the volume **starts increasing** as cracks open. This point is often called the **critical stress**.

4.6 Confinement

- When **lateral confining stress** ($\sigma_2 = \sigma_3$) is applied, the **effective strength increases** and the concrete becomes much more **ductile**. In the slide's curves, the peak stress rises from about 30 MPa (unconfined) to over 150 MPa at 100 MPa confinement, with large strains before failure.
- **Confinement is less effective for higher-strength concrete.**
- This is why **spiral or closely spaced ties** in columns increase their strength and ductility.

4.7 Loading conditions and multiaxial stress

Loading	Failure plane
Uniaxial compression	Cracks **parallel to the load**
Biaxial compression	Splitting along planes **parallel to the free (unloaded) face**; strength 15–25% higher than uniaxial
Compression + tension	Cracks perpendicular to the tensile stress; strength lower
Biaxial tension	Behaves like **uniaxial tension**; failure plane **perpendicular** to the larger tensile stress
Triaxial compression	Very high strength; with high confinement, failure needs very large stress

Mohr–Coulomb theory: although not directly applicable to concrete, it is a **convenient way to represent failure under multiaxial stresses**. Mohr circles are drawn for simple uniaxial tension, compression–tension, simple uniaxial compression and triaxial compression; the **Mohr rupture envelope** is the curve that touches all of them. A stress state whose circle touches the envelope causes failure.

5 Post-peak behaviour and toughness

5.1 Types of post-peak behaviour

Behaviour	After yielding / peak	Example
Strain hardening	Stress keeps **rising** with strain	Mild steel and metals
Elastic–plastic	Stress stays **constant** while strain increases	Idealized steel
Strain softening	Stress **falls gradually** with strain	**Concrete**
Brittle	Stress **drops suddenly** to zero	Glass; concrete tested on a soft machine

5.2 Measuring the post-peak response

- In a **displacement-controlled** machine, **small increments of displacement** are applied, so the **decreasing load after the peak can be recorded**. A load-controlled machine cannot do this: at the peak it keeps pushing, and the specimen breaks suddenly.
- **Strain at failure** (end of the curve) is about **0.005** for normal-strength concrete.
- The **post-peak behaviour depends on the stiffness of the testing machine relative to the specimen**, and on the **strain rate**. A soft machine stores energy and releases it suddenly at the peak, causing explosive failure.
- **Higher-strength concrete is more brittle**: the strain at failure is smaller.

Displacement (closed-loop) control system: a **hydraulic power** unit drives an **actuator** through a **servo-valve**. A **load cell** and a displacement sensor (**LVDT** or strain gauge) send a **feedback** signal to the **controller**, which compares it with the **command signal** from a **function generator** and corrects the servo-valve through an **error signal**. A **data acquisition** system records everything. The goal is a **constantly increasing displacement signal**, so the whole curve, including the falling branch, is captured.

Uses: calculating **fracture toughness** parameters, and representing the **overall ductility** of the system.

5.3 Toughness

- **Toughness = area under the load–deflection curve**, up to a **specified deflection**. It is the **energy absorbed** before failure.
- The specified deflection (end point) **varies between standards**.
- **Compressive, flexural and impact toughness** are the ones usually measured. Toughness matters most for **fibre-reinforced concrete**, which carries load long after cracking.

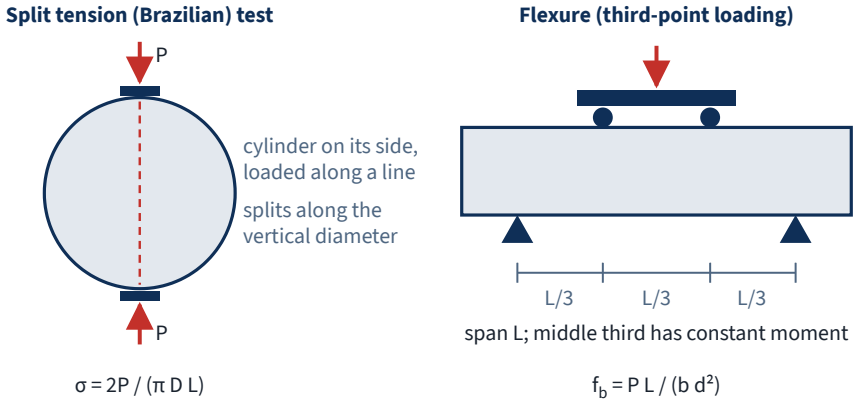
5.4 Flexural toughness test

- A beam on a span of **L = 450 mm**, loaded at **third points** (L/3 spacing).
- **Two LVDTs**, one on each side, measure the **mid-span deflection (δ)**; the average removes twisting effects.
- For fracture tests, a **notched beam** is used, with a **clip gauge** across the notch to measure the **crack mouth opening displacement (CMOD)**, which is used as the control signal.

6 Tensile and flexural strength

6.1 Why tensile strength is measured indirectly

Direct tension tests are hard: gripping the specimen without causing local failure, and keeping the load perfectly axial, is difficult. So **indirect tests** are used.



The two common indirect tension tests.

6.2 Split tensile (Brazilian) test (IS 5816)

- A **cylinder (150 mm × 300 mm) is laid on its side** and loaded along two opposite lines (through thin packing strips) until it **splits along the vertical diameter**.
- Along that diameter, most of the section is under **nearly uniform horizontal tension**, with high compression only very near the loading points. So the cylinder fails in tension.

Split tensile strength

$$\sigma_{max} = \frac{2P}{\pi D L}$$

P = load at failure, D = diameter, L = length of the cylinder (or disc).

Worked example

A 150 × 300 mm cylinder splits at 200 kN. σ = 2 × 200,000 / (π × 150 × 300) = 400,000 / 141,372 = **2.83 MPa**.

6.3 Flexural strength (modulus of rupture), IS 516

- A plain concrete **beam** (150 × 150 × 700 mm on a 600 mm span, or 100 × 100 × 500 mm on a 400 mm span) is loaded at **third points** until it breaks.
- The middle third has **constant bending moment and no shear**, so the beam fails at its weakest section in that zone.

Modulus of rupture

If the fracture is **within the middle third** (a > L/3):

$$f_b = \frac{P\,L}{b\,d^2}$$

If the fracture is **outside the middle third** but a is greater than 133 mm (150 mm beam) or 110 mm (100 mm beam):

$$f_b = \frac{3\,P\,a}{b\,d^2}$$

P = failure load, L = span, b = width, d = depth, a = distance from the line of fracture to the nearer support. If a is smaller than these limits, the result is discarded.

Worked example

A 150 × 150 mm beam on a 600 mm span fails at 25 kN, with the crack in the middle third. $f_b = 25{,}000 \times 600 / (150 \times 150^2)$ = **4.44 MPa**. IS 456 estimate for M30: $f_{cr} = 0.7\sqrt{f_{ck}} = 0.7 \times \sqrt{30}$ = **3.83 MPa**.

6.4 Tensile vs compressive strength

- **Tensile strength is about 8–12% of compressive strength** (often taken as about 10%).
- The ratio **decreases as compressive strength increases**: tensile strength grows more slowly.
- Typical order: **direct tension < split tension < flexural strength (modulus of rupture)**. The slide's graph plots split tensile strength and ¾ of the modulus of

rupture against compressive strength, and both fall on similar curves (so modulus of rupture ≈ 1.33 × split tensile strength).

7 Fatigue strength

Fatigue is failure under **repeated (cyclic) loading** at stresses below the static strength.

7.1 Fatigue load spectrum

Range	Number of cycles	Examples
Low-cycle fatigue	about 1 to 10^3	**Structures subjected to earthquakes**
High-cycle fatigue	about 10^3 to 10^7	**Airport pavements and bridges** (~10^4–10^5); **highway and railway bridges, highway pavements, concrete railroad ties** (~10^5–10^7)
Super-high-cycle fatigue	about 10^7 to 10^9	**Mass rapid transit structures** (~5×10^7); **sea structures** (~5×10^8, from waves)

7.2 S–N (Wöhler) curve

- Fatigue life is shown on an **S–N diagram (Wöhler curve)**: **S_{max}** (maximum stress as a fraction of static strength) on the vertical axis against **N_F** (number of cycles to failure, log scale).
- In the slide's data (S_{min} = 0.035), the points fall on a straight line on a semi-log plot: about **S_{max} = 0.9 at ~10^2 cycles**, falling to about **0.6 at 10^6–10^7 cycles**. Lines for **mean, 5% and 95%** probability of failure are drawn, and **run-outs** (specimens that did not fail) are marked.
- Concrete shows **no clear endurance limit**; a fatigue strength of about **50–55% of static strength at 10^7 cycles** is commonly used.
- Fatigue life increases as S_{max} falls and as the **stress range** ($S_{max} - S_{min}$) narrows.

8 Steel–concrete bond

Bond failure of reinforcing bars happens by **crushing and cracking of the concrete around the bar**, not by the steel itself. With **deformed (ribbed) bars**:

- the **ribs bear against the concrete** and produce **wedging forces**, pushing outwards on the surrounding concrete like a pressurized pipe;
- **inclined cracks develop from the ribs** (at roughly 60° to the bar axis);
- these forces cause **splitting cracks in the cover** along the bar;
- the **confining action of the concrete** (cover thickness, and stirrups) resists splitting.

So bond strength depends on **concrete tensile strength, cover, bar spacing, confinement by stirrups, and bar deformations**. It is measured by **pull-out tests**: a bar embedded in a concrete cube or cylinder is pulled out while bond stress (τ) and **slip** are recorded.

9 Non-destructive testing

9.1 Ultrasonic pulse velocity (UPV) test, IS 13311 (Part 1)

Principle: the **time taken by an ultrasonic pulse to travel through the concrete** is measured; velocity = path length ÷ travel time.

- **Pulse velocity depends on the elastic constants**: modulus **E**, Poisson's ratio **ν** and density **ρ**:

$$V = \sqrt{[\, E\,(1-\nu) \,/\, (\,\rho\,(1+\nu)(1-2\nu)\,)\,]}$$

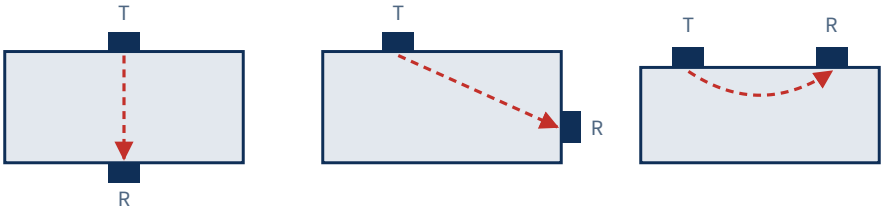
- Dense, sound concrete transmits the pulse fast. **Voids, honeycombing and cracks** make the pulse travel round them (longer path, more time) or reflect it, so the apparent velocity falls.

Factors that influence the reading:

- **contact** between transducer and surface (a couplant such as grease is used),
- **length of the path**,

- **temperature and moisture content** (wet concrete gives higher velocity),
- **presence of reinforcing steel** (steel transmits the pulse faster, so readings near bars are too high).

Arrangements of transducers:



T = transmitter, R = receiver. Direct transmission is the most reliable.

- **(a) Direct**: on **opposite faces**. Maximum energy reaches the receiver; **most accurate**.
- **(b) Semi-direct**: on **adjacent faces**.
- **(c) Indirect (surface)**: both on the **same face**. Least sensitive; used when only one face is accessible (e.g. pavements, walls), and for finding crack depth.

Concrete quality grading by pulse velocity:

Pulse velocity (km/s)	Concrete quality
Above 4.5	**Excellent**
3.5 to 4.5	**Good**
3.0 to 3.5	**Medium**
Below 3.0	**Doubtful**

Worked example

Direct transmission across a 300 mm thick wall gives a travel time of 70 µs. V = 0.300 m / 70 × 10^{-6} s = 4286 m/s = **4.29 km/s → good quality**.

Uses: checking uniformity, locating cracks, voids and honeycombing, monitoring deterioration (e.g. fire or frost damage), and estimating strength with a calibration curve.

9.2 Rebound hammer (Schmidt hammer) test, IS 13311 (Part 2)

Principle:

- A **spring-loaded mass (hammer)** strikes a **plunger** held against the concrete surface with a **defined energy**.
- The hammer **rebounds**, and the **rebound distance** (shown by a **rider** on a **scale**) is read as the **rebound number**.
- The rebound depends on the **surface hardness** of the concrete, which is related to its strength.

Parts: plunger, tubular housing, mass, springs, rider and scale, catch, release button.

Points to remember:

- A **calibration curve** relates rebound number to **compressive strength** (e.g. rebound 20 → about 15 MPa, rebound 35 → about 35 MPa for SSD concrete with limestone aggregate, hammer horizontal).
- The **position of the hammer affects the reading**: separate curves are used for **horizontal**, **vertically downward** and **vertically upward** impacts, because gravity helps or opposes the rebound.
- **Moisture**: **wet** surfaces give **lower** rebound numbers than dry ones for the same strength.
- Other factors: surface smoothness, carbonation (hardens the surface and gives higher readings), aggregate type, age.
- Take at least 9–12 readings in an area and average them. It tests only the **surface layer (about 30 mm)**.

	UPV	Rebound hammer
Measures	Pulse velocity through the whole thickness	Surface hardness

	UPV	Rebound hammer
Finds internal flaws	**Yes** (cracks, voids, honeycombing)	No
Strength estimate	With calibration	With calibration; affected by surface
Speed and cost	Moderate	**Very quick and cheap**

Using both together (the "combined method") gives a better strength estimate than either alone.

Quick revision

- Concrete is multiphase: CA in mortar → sand in paste → C-S-H, CH, pores, unhydrated cement → C-S-H particles and gel pores.
- Aggregate and paste are nearly linear; concrete is non-linear because of ITZ microcracks. E = 30–50 GPa (IS 456: 5000√fck), ν = 0.18–0.24.
- Bond improved by rough aggregate (up to 30% higher tensile/flexural strength), vibration, porous aggregate, limestone.
- Test factors: more end friction → higher load; more slenderness → lower strength; faster loading → higher strength and E; faces must be plane and parallel (capping/grinding).
- Cracking: < 30% negligible; 30–50% bond cracks grow; 50–75% matrix cracks; 75–100% fast growth; post-peak localization. Linear to ~40%; strain at peak ~0.003; at failure ~0.005.
- Higher strength → stiffer, longer linear part, more brittle. Confinement raises strength and ductility (less for HSC). Volumetric strain $\varepsilon 1 + 2\varepsilon 3$.
- Post-peak types: strain hardening (steel), elastic–plastic, strain softening (concrete), brittle. Displacement control needed; toughness = area under load–deflection curve.
- Split tensile $\sigma = 2P/(\pi DL)$. Flexure $f_b = PL/(bd^2)$ (middle third), $3Pa/(bd^2)$ outside. Tensile ≈ 8–12% of compressive.

- Fatigue: low-cycle (earthquakes), high-cycle (bridges, pavements, sleepers), super-high-cycle (MRTS, sea structures). S–N (Wöhler) curve; ~55% at 10^7 cycles.
- Bond failure: rib wedging forces, inclined cracks, splitting of cover, resisted by confinement.
- UPV: V depends on E, ν, ρ; direct/semi-direct/indirect; > 4.5 excellent, 3.5–4.5 good, 3.0–3.5 medium, < 3.0 doubtful. Rebound hammer: surface hardness; depends on hammer position and moisture.

Questions to practise

1. Explain why concrete is called a multiphase composite. Why is its stress–strain curve non-linear even though aggregate and paste are nearly linear? (6 marks)
2. What factors affect the bond between cement paste and aggregate? (4 marks)
3. Discuss the factors that affect the strength of concrete, including Abrams' law and the gel/space ratio. (8 marks)
4. Explain how end restraint, specimen size, loading rate and surface condition affect the measured compressive strength. (8 marks)
5. Sketch the stress–strain curve of concrete in compression and explain the stages of cracking. (8 marks)
6. Explain the effect of confinement and of strength level on the stress–strain behaviour. (5 marks)
7. Distinguish between strain hardening, elastic–plastic, strain softening and brittle behaviour. Why is a displacement-controlled machine needed to record the post-peak response? (6 marks)
8. Describe the split tensile and flexural strength tests. A 150 × 300 mm cylinder fails at 180 kN in the split test; find the tensile strength. (8 marks)
9. What is fatigue? Explain the S–N curve and the fatigue load spectrum for concrete structures. (6 marks)
10. Explain the mechanism of bond failure of deformed bars. (4 marks)

11. Describe the ultrasonic pulse velocity test. What factors affect the results? A pulse takes 85 µs to cross 350 mm of concrete; grade its quality. (8 marks)
12. Describe the rebound hammer test and its limitations. (5 marks)